EE3032 Introduction to VLSI – Midterm

Fall 2023: available latter questions

Source coverage. Only original page 2-2 is available. Questions 5–7 are complete (35 printed points). Questions 1–3 and the opening of Question 4 are unavailable; they are not reconstructed from student answers.

Question 4 continuation (incomplete; opening and parameters missing).
[The source begins mid-sentence:] "voltage of the output V_{out} is 1.2 V and an input stimulus ($0 \rightarrow 1.2$ V) as shown in Fig. 3(b) is applied, how much voltage of the input stimulus (V_{in}) at x is? (4 points) Identify the operation region of the transistors N1, N2, and N3 from t_1 to t_2. (6 points)"

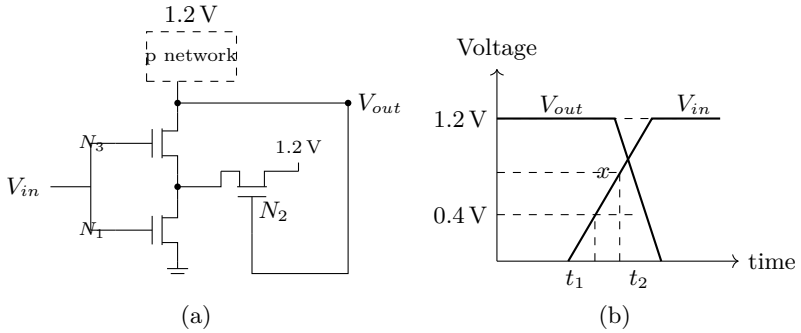


Figure 3: (a) A CMOS logic gate. (b) Waveforms of V_{in} and V_{out}.

5. (10 points) Fig. 4(a) shows a CMOS circuit. Assume that the sizes of NMOS and PMOS transistors in the logic gates are W/L and $2W/L$, respectively. Also, $\mu_n = 2\mu_p$. If the functional inputs of the circuits are $(ABCD) = \{0011, 0101, 1101, 1110\}$, explain the relationship of noise margins NM_L and NM_H (i.e., $NM_L > NM_H$, $NM_L < NM_H$, or $NM_L = NM_H$) of each gate.

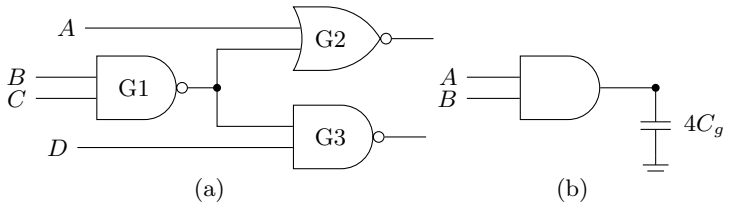


Figure 4: CMOS circuits.

Complete questions (continued)

6. (10 points) For a given technology, the turn-on resistance of a W/L NMOS is R and $\mu_n = 2\mu_p$. Also, the gate capacitance of a W/L NMOS or PMOS transistor is C_g. Fig. 4(b) shows a two-input AND gate drives a $4C_g$ capacitive load. Design the AND gate such that the worst-case delay of the AND gate is smaller than $8RC_g$, where the delay is estimated by using simple RC model and the parasitic capacitance can be ignored. Draw the transistor-level schematic diagram and label the size of each transistor in terms of W/L.

7. (15 points) Realize the following Boolean functions using CMOS logic:

(a) $F = X(Y + Z) + XW$. (5 points)

(b) $F = \overline{X(Y + Z)}$. (5 points)

(c) $F = XY + YZ$. (5 points)

Retained complete-question total: 35 points. The visible Q4 continuation carries 4+6 points, but its missing premise prevents a complete answer. Original Figure 2 belongs to an absent earlier question and is preserved only in the source PDF.